M5. Membership Numbers, Renewal Codes & Nightly Audit

Category B Practice - Problem 5 (Intermediate to Advanced)

Approach

The constructor increments a shared counter and stores the result in a final memberNumber. Renewal codes are checked with character methods and length-first validation. borrowBook(String) reuses borrowBook(); the nightly audit safely skips nulls and uses instanceof to distinguish faculty.

Java Solution

```
class LibraryMember {
    private static int enrolled=0;
    private final String memberNumber;
    protected final int borrowLimit;
    private int booksBorrowed;
    public LibraryMember(int borrowLimit) {
        if(borrowLimit<=0) throw new IllegalArgumentException("Limit must be positive");
        this.borrowLimit=borrowLimit; enrolled++;
        memberNumber="LIB-"+(100+enrolled);
    }
    public String getMemberNumber() { return memberNumber; }
    public void borrowBook() {
        if(booksBorrowed>=borrowLimit) throw new IllegalStateException("Borrow limit reached");
        booksBorrowed++;
    }
    public void borrowBook(String genre) {
        System.out.println("Genre: " + genre);
        borrowBook();
    }
    public int getBooksBorrowed() { return booksBorrowed; }
    public static int getMembersEnrolled() { return enrolled; }
}
class FacultyMember extends LibraryMember {
    private final String department;
    public FacultyMember(int borrowLimit,String department) { super(borrowLimit); this.department=department; }
}
class NightlyAudit {
    static boolean isValidRenewalCode(String code) {
        if(code==null||code.length()!=4) return false;
        return code.charAt(0)=='R' && Character.isDigit(code.charAt(1)) &&
               Character.isDigit(code.charAt(2)) && Character.isUpperCase(code.charAt(3));
    }
    static String processNightlyAudit(LibraryMember[] members) {
        int processed=0, skipped=0, faculty=0, regular=0;
        for(LibraryMember member:members) {
            if(member==null) { skipped++; continue; }
            processed++;
            if(member instanceof FacultyMember) faculty++; else regular++;
        }
        return processed+" processed | "+skipped+" null skipped | "+faculty+" faculty | "+regular+" regular";
    }
    public static void main(String[] args) {
        LibraryMember m=new LibraryMember(3);
        System.out.println(m.getMemberNumber());
        System.out.println(LibraryMember.getMembersEnrolled());
        System.out.println(isValidRenewalCode("R12A"));
        m.borrowBook(); m.borrowBook("Fiction");
        System.out.println(m.getBooksBorrowed());
    }
}
```

Sample Output

```
LIB-101
1
true
Genre: Fiction
2
```

Notes

The field is private and has a getter to satisfy the requirement that it cannot be modified from outside the class.